Westfield Academy

Comprehensive Academic Performance Report

Subject · Sub-Subject · Topic Level Analysis

Student:	Alexandra Chen	**ID:**	STU-20847	**Overall:**	**82% — B-**
Class:	Grade 11 — Advanced Track	**Term:**	Fall Semester 2025	**Attendance:**	96%

Subject Overview

Subject Area	Score	Grade	Sub-subjects Avg Range	Strength
Mathematics	**87%**	**B+**	74% – 97%	*Calculus*
English & Language Arts	**77%**	**C+**	70% – 90%	*Reading Comprehension*
Machine Learning & AI	**90%**	**A-**	82% – 98%	*Supervised Learning*
Science	**75%**	**C**	78% – 82%	*Chemistry*

Mathematics 87% B+

■ Algebra 87% B+

Linear Equations	88%	**B+**	■■■■■■■■■■■■■■■■■
Quadratic Functions	88%	**B+**	■■■■■■■■■■■■■■■■■
Polynomials	83%	**B**	■■■■■■■■■■■■■■■■
Systems of Equations	89%	**B+**	■■■■■■■■■■■■■■■■■

■ Calculus 97% A+

Limits & Continuity	96%	**A**	■■■■■■■■■■■■■■■■■■■
Derivatives	100%	**A+**	■■■■■■■■■■■■■■■■■■■■
Integration	95%	**A**	■■■■■■■■■■■■■■■■■■■
Differential Equations	96%	**A**	■■■■■■■■■■■■■■■■■■■

■ Statistics 88% B+

Probability	76%	**C**	■■■■■■■■■■■■■■■
Distributions	94%	**A**	■■■■■■■■■■■■■■■■■■
Hypothesis Testing	92%	**A-**	■■■■■■■■■■■■■■■■■■
Regression Analysis	91%	**A-**	■■■■■■■■■■■■■■■■■■

■ Geometry 74% C

Euclidean Geometry	67%	**D+**	■■■■■■■■■■■■■
Trigonometry	73%	**C**	■■■■■■■■■■■■■■

Coordinate Geometry	76%	C	
Vectors	81%	B-	

■ Discrete Math — 88% B+

Set Theory	91%	A-	
Logic & Proofs	83%	B	
Graph Theory	89%	B+	
Combinatorics	90%	A-	

English & Language Arts — 77% C+

■ Reading Comprehension — 90% A-

Literal Comprehension	94%	A	
Inferential Reasoning	100%	A+	
Critical Analysis	83%	B	
Text Structure	82%	B-	

■ Writing — 76% C

Argumentative Essays	74%	C	
Narrative Writing	81%	B-	
Grammar & Syntax	77%	C+	
Research Writing	71%	C-	

■ Literature — 70% C-

Poetry Analysis	65%	D	
Fiction & Narrative	81%	B-	
Drama & Theatre	63%	D	
Literary Devices	72%	C-	

■ Communication — 75% C

Public Speaking	67%	D+	
Debate & Rhetoric	80%	B-	
Listening Skills	92%	A-	
Media Literacy	62%	D-	

Machine Learning & AI — 90% A-

■ Foundations	**87%**	**B+**	
Linear Algebra for ML	85%	B	
Probability & Stats	93%	A	
Optimization Theory	90%	A-	
Python Programming	81%	B-	
■ Supervised Learning	**98%**	**A+**	
Linear Regression	97%	A+	
Classification Trees	99%	A+	
Neural Networks	100%	A+	
Support Vector Machines	95%	A	
■ Unsupervised Learning	**87%**	**B+**	
K-Means Clustering	82%	B-	
Dimensionality Reduction	94%	A	
Autoencoders	86%	B	
Anomaly Detection	87%	B+	
■ Deep Learning	**82%**	**B-**	
Convolutional Networks	79%	C+	
Recurrent Networks	82%	B-	
Transformers	93%	A	
Generative Models	73%	C	
■ Applied AI	**84%**	**B**	
Natural Language Processing	85%	B	
Computer Vision	93%	A	
Reinforcement Learning	87%	B+	
Model Evaluation	73%	C	

Science			75% C
■ **Biology**		78%	C+
Cell Biology	71%	C-	
Genetics & DNA	68%	D+	
Ecology	85%	B	
Human Physiology	86%	B	
■ **Chemistry**		82%	B-
Atomic Structure	78%	C+	
Chemical Bonding	79%	C+	
Organic Chemistry	89%	B+	
Thermodynamics	81%	B-	
■ **Physics**		80%	B-
Mechanics	82%	B-	
Electromagnetism	65%	D	
Quantum Physics	88%	B+	
Optics & Waves	86%	B	

Cross-Subject Concept Connections

Topic A		Topic B	Connection
Statistics (Math)	↔	Probability & Stats (ML)	*Shared core: probability distributions, hypothesis testing, Bayesian reasoning*
Linear Algebra (Math)	↔	Linear Algebra for ML (ML)	*Foundation for neural network weights, PCA, and matrix operations*
Algebra — Regression (Math)	↔	Linear Regression (ML)	*Same mathematical model; ML extends with gradient descent & regularization*
Logic & Proofs (Discrete Math)	↔	Optimization Theory (ML)	*Formal reasoning underpins loss function design and convergence proofs*
Graph Theory (Discrete Math)	↔	Neural Networks (ML)	*Graphs model computation: neurons are nodes, weights are edges*
Argumentative Writing (English)	↔	Model Evaluation (ML)	*Both require evidence-based reasoning, structured claims, and counterarguments*
Genetics & DNA (Biology)	↔	Generative Models (ML)	*Both encode, replicate, and vary information — structural analogy*
Electromagnetism (Physics)	↔	Transformers (ML)	*Attention mechanisms echo field interactions — signal propagation metaphors*

Advisor: Dr. Patricia Moore ________________ Principal: ________________ Parent/Guardian: ________________